Case Study 1: National Hospital Management System

Background:

A national public hospital plans to develop an integrated **Hospital Management System (HMS)** to manage patient records, appointments, billing, laboratory results, and staff schedules. The system will be used across multiple hospitals and must comply with **government healthcare regulations** and **data privacy laws**.

Project Characteristics:

- Requirements are **clearly defined and documented** by the Ministry of Health
- Changes to requirements are **rare and strictly controlled**
- The system must meet **high safety and reliability standards**
- Extensive **documentation and formal approvals** are required
- Testing and validation must be traceable to requirements
- Customer (hospital administration) involvement is **limited after requirements approval**

Constraints:

- Failure or incorrect behavior may lead to **serious consequences**
- Regulatory audits are expected
- Development timelines are fixed and contract-based

Task :

1. Which **software process model** is most appropriate for this project?
2. Why does this model fit the project's **regulatory and safety requirements**?
3. Why would **Agile-based models** be less suitable in this scenario?

Case Study 2: Startup Mobile Fitness Application

Background:

A technology startup is developing a **mobile fitness application** that tracks workouts, diet plans, and daily activity. The startup aims to release a **minimum viable product (MVP)** quickly to test the market and attract investors.

Project Characteristics:

- Requirements are **uncertain and frequently changing**
- Features evolve based on **user feedback**
- Time-to-market is critical
- The customer (startup founders) is **highly involved**
- The development team is small and cross-functional
- Frequent updates and enhancements are expected

Constraints:

- Limited budget
- Competitive market pressure
- Need for rapid feature experimentation

Task:

1. Which **software process model** is best suited for this project?
2. How does the chosen model handle **changing requirements**?
3. What risks might arise if a **plan-driven model** were used instead?

Case Study 3: Autonomous Vehicle Control Software

Background:

An automotive company is developing **control software for an autonomous vehicle system**. The software controls braking, steering, and obstacle detection. The system must comply with **international automotive safety standards**.

Project Characteristics:

- Requirements are **strict, formal, and traceable**
- Extensive **verification and validation** are mandatory
- Testing must be planned **in parallel with development**
- Any defect may lead to **severe safety risks**
- Changes are allowed but are **costly and heavily reviewed**
- Development involves large, specialized teams

Constraints:

- Compliance with ISO and automotive standards
- High cost of failure
- Long-term maintenance requirements

Task for Students:

1. Which software process model ensures **strong traceability** between requirements and testing?
2. Why is **early test planning** critical in this system?
3. Compare this model briefly with the Waterfall model in this context.

Case Study 4: University E-Learning Platform Enhancement

Background:

A university wants to enhance its existing **e-learning platform** by adding features such as live quizzes, discussion boards, and analytics dashboards. The system is already in use by thousands of students and instructors.

Project Characteristics:

- The system is developed in **phases**
- New features are added gradually
- Existing functionality must remain stable
- Feedback from instructors and students is collected after each release
- The university wants **early delivery of key features**

Constraints:

- Limited development staff
- Continuous system availability required
- Budget constraints

Task for Students:

1. Which process model supports **gradual feature delivery**?
2. Why is this model suitable for extending an existing system?
3. What challenges could arise if all features were developed at once?